Divergent Hypothesis Dynamics

Controlled Exploratory Cognition over Heterogeneous Model Spaces

DogeOmegaRACLE Project

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Abstract

Divergent Hypothesis Dynamics (DHD) is a mathematical framework for exploratory cognition: the deliberate generation, mutation, retention, and provenance tracking of a heterogeneous population of executable hypotheses without treating generation as belief. The framework is motivated by nonstationary and reflexive environments in which a conventional inference system can become locally coherent yet globally myopic because the useful explanatory model lies outside its currently favored family. DHD therefore separates model-space exploration from epistemic adjudication. A hypothesis is a typed executable object in a coproduct of model families; generators are typed partial stochastic maps; comparison is multi-view rather than a universal scalar distance; and population control is performed by a novelty/coverage archive under explicit compute, memory, and latency budgets. The system is externally metacontrolled by Psychotic Decision Calculus (PDC) through a control tuple that independently specifies generator allocation, mutation magnitude, resource budget, and archive breadth. DHD itself does not assign posterior belief, select trades, size positions, or execute actions. We formalize contradiction between hypotheses without permitting internal incoherence within a hypothesis, distinguish layered notions of equivalence, define a labeled provenance DAG, and specify a reflexive model family using dynamic/performative semantics rather than static DAG cycles. We prove elementary archive-boundedness and type-closure results, import the classical greedy guarantee for monotone submodular selection under its stated assumptions, and show why determinant-based selection does not by itself imply observational-equivalence compression. The central empirical claim remains deliberately unproved: controlled structural diversity may reduce validated discovery time relative to matched-compute baselines. We provide a falsifiable staged validation program, ablations, and kill criteria. No empirical market-performance claim is made.

1. Introduction

A predictive system can fail in two different ways. It can reason badly inside the correct model class, or it can reason well inside the wrong model class. The second failure is more difficult because increasing inference quality can strengthen commitment to a misspecified explanatory family. In nonstationary, partially observed, adversarial, or reflexive environments, the relevant causal structure, temporal scale, latent regime, or entire model family may change faster than a conventional estimator can recover.

Divergent Hypothesis Dynamics (DHD) addresses that failure mode by making **exploratory model generation** a first-class computational process. Its organizing principle is:

Generating a hypothesis is not believing a hypothesis.

The distinction is stronger than ordinary Bayesian model uncertainty. Bayesian model averaging, sequential Monte Carlo, and interacting-model filters maintain alternatives in order to approximate or combine beliefs over a declared family. DHD instead maintains alternatives because some useful explanations may initially have negligible support, live in a different model family, or be generated by a structural transformation not represented in the current posterior. The nearest established ideas are novelty search and quality-diversity methods [1], [2], [3], population methods [4], [5], causal model search [6], [7], probabilistic programming [8], determinantal and submodular subset selection [9], [10], and performative prediction [11]. DHD does not claim novelty for those components individually.

The proposed contribution is architectural. DHD combines a typed heterogeneous hypothesis space, an explicitly controlled family of structural generators, a bounded exploratory archive, a first-class provenance genealogy, and a strict separation between generation and later epistemic adjudication. The control policy is supplied by PDC, whose Version 2.1 specification is itself cognition/meta-control only and does not construct or execute trades [12]. DHD therefore complements PDC rather than replacing it.

This paper is a governing theoretical specification, not a market-results paper. It formalizes the mathematical core that survived two rounds of adversarial internal research. Claims that were not verified are labeled as conjectures, engineering heuristics, or empirical hypotheses rather than promoted to theorem status.

1.1 Scope and non-scope

DHD owns hypothesis generation, mutation, exploratory archive maintenance, contradiction-preserving representation, provenance, and exploration diagnostics. It does **not** own posterior belief, final forecast selection, action valuation, trading policy, instrument choice, position size, orders, or execution. Its output is an epistemically raw but structurally organized population for a later adjudicator.

The framework makes no clinical or psychiatric claim. The historical DogeOmegaRACLE project explored unusual cognition as a design metaphor, but DHD is defined entirely in computational terms: search breadth, operator structure, model-family transitions, causal alternatives, temporal reinterpretation, reflexivity, and resource allocation.

1.2 Central scientific hypothesis

Empirical Hypothesis 1 (Controlled-diversity advantage). There exist environments in which, under matched compute, memory, latency, and model-family access, a controlled DHD process produces a hypothesis that later passes pre-registered adjudication earlier than random valid search, conventional model averaging, or a non-metacontrolled novelty-search baseline.

This is the core proposition that justifies DHD as an independent framework. It is not assumed true. Section 14 defines experiments that can falsify it.

2. Mathematical objects and notation

Let time be indexed by ticks $t \in N$. DHD maintains a logical population of hypotheses and several physical stores. Let K be a finite or countable set of model-family tags. Examples include state-space models, structural causal models, regime-switching models, probabilistic programs, agent-based models, and reflexive dynamic models.

Definition 1 (Typed hypothesis space). The ambient hypothesis space is the disjoint union

$$H = \coprod_{k \in K} H_k .$$

An element $h \in H$ has exactly one family tag $k(h)$ even when two families can express observationally equivalent behavior. The coproduct preserves family identity and prevents parameters from unrelated model classes from being treated as coordinates in one Euclidean vector space.

Definition 2 (Hypothesis object). A DHD hypothesis is the typed object

$$h = (k, M, V, A, I, T, \Phi, \Omega, \Pi),$$

where:

Component	Meaning
k	model-family tag
M	executable generative model or program
V	typed variables and domains
A	explicit assumptions and scope conditions
I	intervention/counterfactual semantics when defined
T	temporal semantics: lags, horizon, memory, aggregation
Phi	testable or conditionally testable implications
Omega	predictive-consequence operator
Pi	provenance record and genealogy reference

Not every family must implement every operation. A noncausal forecasting model can leave I undefined; a static causal model can have trivial temporal semantics; an underidentified model can declare its identification status explicitly.

Definition 3 (Minimum family interface). Every admitted family supports the following partial interface:

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GenerateObservables(h,                                          context)
PredictDistribution(h,                    horizon,              context)
Validate(h)
Complexity(h)
Provenance(h)
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with optional Intervene(h, intervention) and Counterfactual(h, evidence, action) functions when the family defines those semantics.

Assumption 1 (Executable consequences). Every hypothesis eligible for the active archive can generate at least one non-vacuous observable or interventionally meaningful consequence over a declared context and horizon.

This assumption is what makes cross-family comparison possible without inventing a universal parameter geometry.

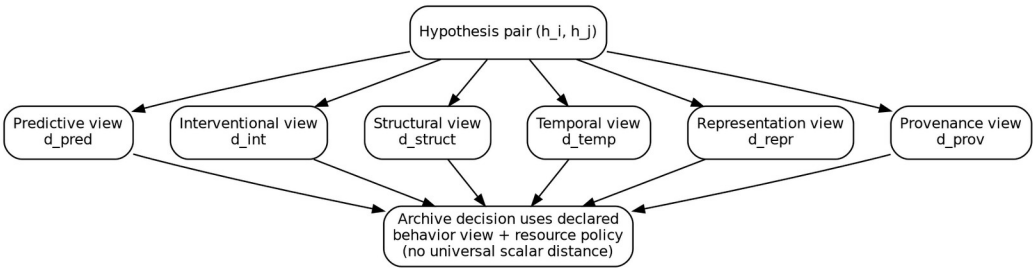
3. Multi-view hypothesis geometry

A global scalar distance between arbitrary hypotheses is neither required nor generally well motivated. Predictive distributions, causal graphs, temporal assumptions, model-family identity, and provenance are different objects with different invariances. DHD therefore uses a profile of view-specific comparisons rather than an arbitrary weighted sum.

Definition 4 (Multi-view distance profile). For two hypotheses h_i, h_j define

$$d(h_i, h_j) = (d_{pred}, d_{int}, d_{struct}, d_{temp}, d_{repr}, d_{prov}),$$

where a component may be undefined if the relevant semantics do not exist for both members.



Multi-view comparison of two hypotheses. The archive uses declared behavior-space views rather than an unconditional scalar distance.

3.1 Predictive view

Let $Q_i^{c,\tau}$ and $Q_j^{c,\tau}$ denote the predictive distributions produced under the same context c and forecast horizon τ. Suitable comparisons include Wasserstein distance when outcome geometry matters [13], Jensen-Shannon distance when densities are available, and maximum mean discrepancy (MMD) when only samples are available [14].

For a characteristic kernel k on observable space, squared MMD is

$$MMD_k^2(P, Q) = E_{X,X' \sim P} k(X, X') + E_{Y,Y' \sim Q} k(Y, Y') - 2\, E_{X \sim P, Y \sim Q} k(X, Y).$$

The predictive view is the default cross-family comparison because Assumption 1 requires each active hypothesis to expose consequences.

3.2 Causal and interventional views

When both hypotheses define a common intervention set J, define an interventional response discrepancy

$$d_{int}(h_i, h_j) = \int_J D\big(P_i(Y \mid do(j)), P_j(Y \mid do(j))\big)\, d\mu_J(j),$$

for a declared distributional discrepancy D and reference measure μ_J. For DAG-based causal models, Structural Intervention Distance can be used as a structural/interventional diagnostic under its stated assumptions [15]. It is not a universal metric on all of H.

3.3 Kernel caution

A common temptation is to transform any distance d into

$$k(x,y)=\exp\left(-\frac{d(x,y)^2}{2\sigma^2}\right).$$

This is not automatically positive semidefinite. Schoenberg's results require the relevant distance structure to satisfy negative-type conditions [16], [17]. DHD therefore permits a kernel only on a channel for which positive semidefiniteness is established. MMD is constructed in an RKHS and does not require this conversion.

Engineering Rule 1. No implementation may create a DPP, kernel ridge model, or repulsive particle rule from a similarity matrix unless the required positive-semidefinite conditions are verified or an indefinite-kernel method is explicitly used.

4. Equivalence, redundancy, and contradiction

DHD is intended to maintain contradictory explanations without confusing contradiction with invalidity. It must also avoid wasting resources on purely representational duplicates. These goals require a hierarchy of relations rather than one equivalence class.

4.1 Layered equivalence

Definition 5 (Equivalence hierarchy). DHD distinguishes at least the following relations:

$$\sim_{syn},\sim_{par},\sim_{obs},\sim_{Markov},\sim_{int},\sim_{cf},\sim_{task}.$$

They mean syntactic, reparameterization, observational, Markov, interventional, counterfactual, and task/horizon equivalence, respectively. These relations are not interchangeable. For example, two Markov-equivalent DAGs encode the same conditional-independence structure under the relevant assumptions but can remain distinct as causal hypotheses once intervention semantics are considered [6], [7].

Resource compression should therefore occur at a declared layer. An archive may keep one representative per observational-equivalence class for a predictive task while preserving structurally distinct representatives in a compressed index when later interventions can distinguish them.

4.2 Contradiction between hypotheses

Definition 6 (Predictive contradiction). For event E, tolerance $\epsilon \in (0,1/2)$, and common context c, h_i and h_j are predictively contradictory if

$$P_i(E \mid c)\geq 1-\epsilon \text{ and } P_j(E \mid c)\leq \epsilon,$$

or vice versa.

Analogous contradiction relations can be declared for causal sign, regime identity, temporal lag, or intervention response. A pair can be contradictory in one view and similar in another.

Proposition 1 (Contradiction isolation). Suppose each $h \in H$ is validated independently and DHD never forms the classical conjunction $\bigwedge_{h\in A_t} h$ as a single theory. Then contradiction between distinct members of A_t does not imply internal inconsistency of any member.

Proof. Hard validity is evaluated on each typed hypothesis object. Contradiction is a binary relation between objects. Since DHD stores the objects as alternatives rather than conjoining their propositions into one deductive theory, a contradiction $h_i \perp h_j$ does not alter the validation result for h_i or h_j. This is the same architectural separation exploited by assumption-based truth-maintenance systems, which maintain alternative consistent environments rather than allowing one contradiction to collapse the whole knowledge base [18]. ▫

This result is simple but central: DHD is allowed to be collectively contradictory while each executable member remains internally coherent.

5. Divergent generation operators

A generator is not a free-form text prompt. It is a typed transformation with declared domain, codomain, preconditions, stochasticity, cost, and provenance effect.

Definition 7 (Generator). A generator G_g is a typed partial stochastic map

$$G_g : H_{in} \times E_t \times U_t^D \times \Xi_g \dashrightarrow P\left(H_{out}\right),$$

where E_t is generation context, U_t^D is the DHD control state, Ξ_g contains operator parameters, and P denotes a probability law over child hypotheses. The partial arrow denotes permitted failure when preconditions are not met.

5.1 Core generator classes

The minimum viable theory uses five broad classes; suboperators retain distinct provenance where useful.

Generator class	Typical transformation	Primary divergence channel	Example failure
G_local	parameter / mechanism perturbation	parameter, predictive	scale too small to produce novelty
G_reg	split/merge regime; change lag/horizon	regime, temporal	label proliferation / aliasing
G_causal	add/remove/reverse edge; add confounder; counterfactual surgery	causal, interventional	unsupported identification assumptions
G_class	change entire model family	representational, predictive	incomparable outputs without shared consequence API
G_refl	add belief-action-state feedback semantics	reflexive, temporal	unstable or non-executable recursion

Optional extensions include $G_{analogy}$, $G_{adversarial}$, and a carefully defined rare-region generator. Causal inversion is best treated as a provenance-distinct special case of G_{causal}, not as evidence that the operator family must be mathematically disjoint.

Definition 8 (Channel-coverage matrix). Let C be the declared set of divergence channels. Define

$$A_{g,c} = 1\{G_g \text{ can alter channel } c\}.$$

A generator library is channel-complete for C when every declared channel has at least one generator. This is a design invariant, not a proof of global search completeness.

5.2 Operator composition and path dependence

For generators G_a, G_b, composition need not commute:

$$G_b \circ G_a(h) \not\equiv_d G_a \circ G_b(h).$$

For example, adding a latent confounder before reversing an edge can create a different admissible causal family from reversing the edge first and then introducing the confounder. Operator-path dependence is therefore a legitimate source of exploratory divergence and must be recorded in provenance.

Definition 9 (Reachable set). Given seed set H_0 and generator family G,

$$Reach(H_0, G) = \overline{\{G_{i_m} \circ \cdots \circ G_{i_1}(h) : h \in H_0, m < \infty, \text{all steps valid}\}}.$$

No claim of global ergodicity is made. The relevant engineering question is whether the generator family provides useful coverage of a declared reachable subspace under finite resources.

6. Controlled instability and the PDC interface

The original scalar control $\rho^D \in [0,1)$ is rejected because it conflates distinct physical decisions: which generator to use, how large a mutation to make, how many resources to allocate, and how broad the archive should remain.

Definition 10 (DHD control tuple). PDC supplies

$$u_t^D = \left(\pi_t^G(\cdot \mid C_t^{PDC}), s_t, B_t, \ell_t\right),$$

where:

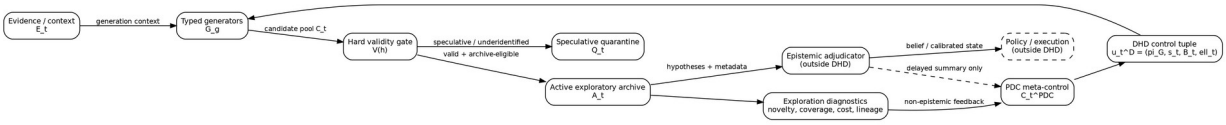
π_t^G is a probability distribution over generator classes;

s_t is a vector of operator-specific mutation scales;

$B_t = \left(B_t^{cpu}, B_t^{mem}, B_t^{lat}, B_t^{pop}\right)$ is a resource budget;

ℓ_t is an archive novelty/admission scale.

The components are independent actuators. A system may, for example, keep a narrow archive while allowing occasional large causal mutations, or maintain broad behavioral coverage while restricting every individual mutation to be local.



DHD architecture and authority boundaries. DHD generates and archives; PDC allocates exploratory cognition; epistemic adjudication and policy remain separate.

6.1 Firewalled operator adaptation

Operator allocation may adapt to DHD-internal rewards such as accepted novelty, new coverage, lineage diversity, validity rate, and cost efficiency. It must not be updated synchronously from realized predictive correctness, posterior belief, or trade profit; doing so would collapse generation into epistemic optimization.

Delayed adjudicator feedback may be used by a higher-level research or configuration process, but the interface must label it as external performance feedback rather than as the DHD definition of novelty.

6.2 No lambda or kappa control inside DHD

PDC Version 2.1 retains λ only as a terminal compatibility projection and removes κ from causal decision logic [12]. DHD therefore does not use historical λ or κ as internal controls. Any future integration paper may define observational relationships, but DHD's canonical actuators are Equation (14).

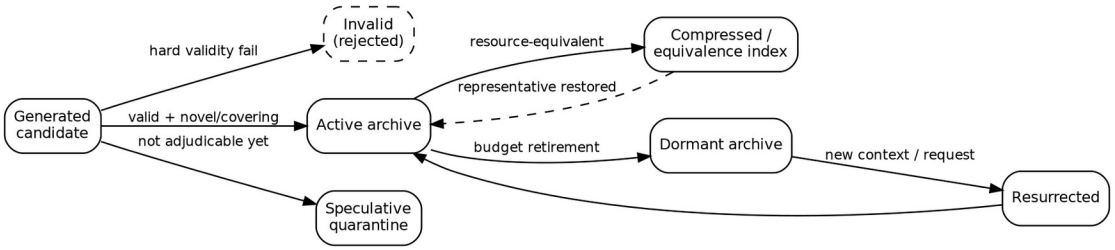
7. Archive and population dynamics

DHD's logical population is not a posterior particle cloud. The system distinguishes candidates, active members, dormant members, speculative objects, and compressed equivalence representatives.

Definition 11 (Archive state). At tick t define

$$S_t = \left(C_t, A_t, D_t, Q_t, E_t, G_t^H\right),$$

where C_t is the candidate pool, A_t the active archive, D_t the dormant archive, Q_t the speculative quarantine, E_t the equivalence/compression index, and G_t^H the provenance graph.



Lifecycle of a generated hypothesis. Epistemic rejection is not an archive transition owned by DHD.

7.1 Candidate generation and hard validity

From a parent set $P_t \subseteq A_t \cup D_t$, DHD draws generators according to π_t^G and produces

$$C_{t+1} = \bigcup_{h \in P_t} \bigcup_{m=1}^{M_t(h)} G_{g_{h,m}}\left(h; E_t, u_t^D, \xi_{h,m}\right).$$

A candidate is active-eligible only if it passes the hard validity predicate $V(h)=1$.

Definition 12 (Hard validity). The predicate requires:

syntactic and type validity for the declared family;

numerical executability to a declared tolerance;

internal causal/dynamical consistency appropriate to the family;

at least one non-vacuous observable or interventional consequence;

complete provenance sufficient for reproduction;

resource feasibility for the current evaluation tier.

Low current prior probability, poor current fit, unusual parameter values, or unconventional structure are not hard-invalidating conditions. Those quantities belong to later adjudication or are recorded as metadata.

7.2 Testability classes

Candidates are labeled:

T1: executable and currently testable;

T2: conditionally or eventually testable;

T3: currently underidentified but with declared discriminating evidence;

T4: speculative/unadjudicable;

T5: invalid.

Only T1-T3 count toward active archive metrics. T4 can be retained in Q_t with provenance but does not consume the active archive budget. T5 is rejected.

7.3 Behavior-space novelty and coverage

A universal notion of "coverage of all hypotheses" is undefined because there is no canonical probability measure on all valid models. DHD therefore declares a behavior descriptor map

$$\beta_t : H \to B_t,$$

where (B_t, d_B) is a task-specific metric descriptor space constructed from predictive or interventional consequences over a fixed probe set.

Definition 13 (Behavioral novelty). Relative to archive A,

$$v(h \mid A) = \begin{cases} \min_{a \in A} d_B(\beta(h), \beta(a)), & A \neq \varnothing, \\ +\infty, & A = \varnothing. \end{cases}$$

Definition 14 (Coverage under a declared measure). For reference region B^{ref} with finite measure μ_B,

$$Cov_\epsilon(A) = \frac{\mu_B\left(\bigcup_{h \in A} B_\epsilon(\beta(h)) \cap B^{ref}\right)}{\mu_B(B^{ref})}.$$

When no defensible reference measure exists, DHD reports occupied descriptor cells, packing statistics, or nearest-neighbor summaries rather than claiming probability-mass coverage.

7.4 Minimal archive rule

Let $V_{t+1} = \{h \in C_{t+1} : V(h) = 1, class(h) \in \{T1, T2, T3\}\}$. A minimal threshold archive admits h if it creates an empty behavior region or satisfies $v(h \mid A_t) \geq \ell_t$. If the resulting archive exceeds the budget, a declared budgeted selection function is applied.

A useful coverage objective is facility location:

$$F(S) = \sum_{u \in U} \max_{h \in S} s(u, h),$$

where U is a finite set of probe cells/scenarios and s is a bounded nonnegative coverage similarity.

Theorem 1 (Classical greedy submodular guarantee). If F is normalized, monotone, and submodular and selection is constrained by $|S| \leq K$, the standard greedy algorithm returns S_g satisfying

$$F(S_g) \geq (1 - 1/e) F(S^\iota),$$

where S^ι is the optimal cardinality-K subset [10].

This theorem applies only when the selected archive objective actually satisfies those conditions. It does not imply that DHD's entire multi-view population problem is submodular.

7.5 Packing bound

Proposition 2 (Archive cardinality under ϵ-separation). Let $\beta(A_t) \subset B$ be ϵ-separated and let B^{reach} be totally bounded. Then

$$|A_t| \leq M(\epsilon; B^{reach}),$$

where $M(\epsilon; B^{reach})$ is the ϵ-packing number.

Proof. An ϵ-separated set contains at most one member per point of a maximal ϵ-packing. By definition the cardinality of any such set is bounded by the packing number. ▫

The proposition converts combinatorial explosion into an explicit dependence on the chosen behavior space, reachability region, and novelty scale. It is not a universal finite bound: total boundedness is an assumption that must be checked or enforced by truncation.

8. Determinantal point processes: optional, not canonical

DPPs are useful models of repulsive subset selection [9], but Round-1 research overclaimed their role. For a positive semidefinite L-ensemble, a subset Y has probability proportional to

$$P(Y) \propto det(L_Y).$$

The determinant rewards diversity in the feature geometry encoded by L. Exact MAP subset selection is combinatorial, and practical methods are approximate [19].

Proposition 3 (Cardinality is imposed, not discovered). If a DPP-based selector solves

$$\max_{Y \subseteq C} \det\left(L_Y\right) \text{ subject to } \left|Y\right| \le K,$$

then $\left|Y\right| \le K$ because of the explicit constraint, independent of the determinant objective.

Proposition 4 (No automatic observational compression). Determinant maximization does not imply removal of observationally equivalent hypotheses unless the kernel represents observational equivalence.

Proof by counterexample. Let $h_1 \sim_{obs} h_2$ but let their feature vectors used by L be orthogonal because the kernel encodes structural or provenance differences. Then $L_{12} = 0$ and $\det\left(L_{\{1,2\}}\right) = L_{11} L_{22} > 0$, so both can be favored simultaneously. Therefore observational equivalence is not compressed unless the kernel or a separate equivalence rule makes such members maximally similar. ▫

DPPs are consequently optional candidate-subset mechanisms inside the broader archive architecture, not the definition of DHD.

9. Provenance and genealogy

DHD must be able to prove when and how a hypothesis was created. Provenance is therefore part of the mathematical object rather than a debug log.

Definition 15 (Provenance DAG). Let

$$G_t^H = \left(V_t^H, E_t^H\right)$$

be a directed acyclic graph whose nodes are immutable hypothesis versions and whose labeled edges have the form

$$h_p \xrightarrow[\text{evidence hash, control, RNG, code version}]{g, \xi_g} h_c.$$

This extends standard data-provenance ideas [20] to model-generation lineage.

Verified uses of G_t^H include reproducibility, lineage-aware novelty diagnostics, operator credit assignment using internal DHD rewards, auditability, and deduplication. A seed, parent content hash, generator version, parameters, evidence snapshot hash, and random-number state should be sufficient to reproduce a deterministic child-generation run within declared numerical tolerance.

9.1 Provenance does not solve multiplicity

Adaptive DHD search can produce thousands of hypotheses from shared data. Shared ancestry does not itself specify the dependence among later statistical tests. The number of lineage roots is therefore not a valid substitute for the number or dependence structure of hypotheses in family-wise error or false-discovery control.

DHD supplies provenance metadata to the adjudicator; the adjudicator must use a statistical procedure appropriate to adaptive or sequential testing. Classical FDR control [21] and sequential ideas such as alpha-investing [22] are relevant starting points, but this paper does not claim a complete solution for adaptive hypothesis generation.

10. Reflexive hypothesis family

Markets and other strategic environments can be endogenous: forecasts or publicly visible signals can change behavior, which changes future data. A static DAG is not sufficient for a state-belief-action-state feedback loop. DHD therefore defines a separate reflexive family using dynamic/performative semantics [11].

Definition 16 (Reflexive model family). A member of H_{refl} specifies

$$\begin{aligned} X_{t+1} &= f(X_t, U_t, \varepsilon_{t+1}), \\ B_t &= \mathscr{B}(X_{\le t}, Y_{\le t}, M_t), \\ U_t &= \pi(B_t, M_t), \\ M_{t+1} &= \mathscr{M}(M_t, \hat{X}_t, U_t). \end{aligned}$$

where X_t is environment state, B_t participant belief state, U_t participant action, and M_t public information/model exposure. These variables describe the hypothesis's modeled agents; they are not DogeOmegaRACLE trade instructions.

A reflexive candidate must state either conditions under which the recursion is well posed (for example, contraction, bounded sensitivity, stable equilibrium, or finite-horizon simulation semantics) or an explicit nonconvergence behavior it predicts. G_{refl} generates members of this family; the adjudicator later evaluates whether the hypothesized feedback exists.

11. Rare-region, analogical, and abductive generation

11.1 Rare-region generation

The phrase $P(h) \ll 1$ is undefined unless a probability measure over hypotheses has been declared. DHD therefore distinguishes:

low prior probability under a specified prior $\pi(h)$;

low proposal probability under the current generator policy;

tail parameter regions under a family-specific reference measure;

low-frequency latent regimes;

rare predicted events under Q_h.

A rare-region generator must state which reference measure defines rarity. "Unusual" is not a mathematical operator.

11.2 Analogical generation

Analogical generation is optional because semantic analogy can easily become decorative language. If retained, it must output a typed structure-mapping record inspired by relational analogy theory [23]: source model, target domain, mapping, preserved relations, broken relations, induced hypothesis, and at least one discriminating consequence. Otherwise the object is not admitted as a scientific hypothesis.

11.3 Abduction

DHD's broad function is compatible with abductive generation—constructing candidate explanations for observed phenomena—but DHD does not equate explanatory adequacy with truth. A candidate explanation still requires later evidence-based adjudication. Probabilistic programming is attractive because it makes many such candidates executable [8].

12. Resource model and computational complexity

DHD is explicitly resource bounded. Let the cost vector of operation o be

$$c(o) = (c_{cpu}, c_{mem}, c_{lat}, c_{sim}, c_{store}).$$

Budgets are vector-valued. They should not be collapsed into a scalar unless a declared exchange-rate model exists.

For N active hypotheses, naive pairwise comparison is $O(N^2 C_d)$ where C_d is the cost of the selected view. Expensive model simulation can dominate the nominal quadratic factor. DHD should therefore use a staged pipeline:

canonical hashes and type-aware normal forms for exact duplicates;

cheap behavior sketches for approximate neighborhood search;

approximate nearest-neighbor novelty estimation;

expensive predictive/interventional comparisons only within local neighborhoods;

archive selection on a coreset, descriptor cells, or bounded candidate batch.

Sequential Monte Carlo provides a useful warning: posterior resampling can lead to particle impoverishment and genealogical degeneracy [4], [5]. DHD imports the lesson that diversity requires active preservation, but DHD is not itself an SMC posterior approximation, so SMC convergence results do not transfer wholesale.

13. DHD output contracts

13.1 DHD to PDC

DHD returns only exploratory diagnostics and state:

$$R_t^D = \left(A_t, D_t, m_t^{div}, m_t^{cov}, m_t^{valid}, m_t^{cost}, G_t^H \right).$$

PDC may use these diagnostics to adjust future u_t^D. DHD must not return a trade direction, order size, execution command, or belief weight masquerading as an exploration metric.

13.2 DHD to epistemic adjudicator

Every active hypothesis exports at least:

immutable hypothesis ID and family tag;

executable model/version;

predictive-consequence interface;

intervention/counterfactual interface when defined;

temporal horizon and assumptions;

testable implications and identification status;

provenance DAG reference;

generation/resource cost;

proposal or prior metadata when those measures exist;

equivalence-class metadata.

DHD does not export a DHD-level “confidence” score. A model may have internal uncertainty parameters, but those are properties of M, not DHD belief.

14. Validation, ablation, and falsification

The framework must be validated in worlds where the truth is known before it is evaluated on historical markets.

14.1 Stage 1: synthetic discovery

The initial suite contains at least:

Regime-switching world: a local baseline is initially correct, followed by a hidden structural transition.

Causal-reversal world: $X \to Y$, $Y \to X$, and $X \leftarrow Z \to Y$ alternatives are distinguishable under designed interventions.

Observational-equivalence world: models agree observationally but diverge under intervention.

Rare-regime world: a low-frequency but valid structural regime appears.

Temporal-aliasing world: similar correlations arise from distinct lag structures.

Reflexive world: participant belief/action feedback changes the data-generating process.

Adversarial-contamination world: observations are selectively corrupted.

Model-class-misspecification world: the true generator is outside the baseline family.

Null world: structural divergence confers no advantage.

The null world is mandatory. A framework that cannot fail under a world designed to contain no benefit is not being falsified.

14.2 Baselines

Compute-matched comparisons include random valid hypothesis generation, Bayesian model averaging or stacking, SMC/IMM where applicable, ordinary novelty search, MAP-Elites or another archive search with the same model access, causal-discovery baselines, the same generators without PDC metacontrol, and an LLM/program generator without DHD archive logic.

Budgets should be matched on model evaluations, simulation calls, memory, and wall-clock time where practical.

14.3 Discovery-time metrics

Raw generation time rewards systems that enumerate everything. DHD therefore records separate timestamps:

$$t_{gen}, t_{reg}, t_{sup}, t_{pass}, t_{event}.$$

These denote generation, immutable prospective registration, first supporting evidence, adjudication pass, and event occurrence. The primary lead-time metric is validated lead

$$\Delta t_{valid} = t_{pass}^{baseline} - t_{pass}^{DHD}.$$

Generation lead is reported only as a secondary metric for hypotheses that later pass pre-registered adjudication. Precision/recall over generated candidates, false-discovery burden, and compute cost must be reported alongside lead time. A system that emits every possible model at $t = 0$ should therefore perform poorly on efficiency and multiplicity even if one emitted hypothesis happens to be correct.

14.4 Ablations

Required ablations include removing each generator class, forcing local-only mutation, forcing maximal structural mutation, disabling equivalence compression, holding population size fixed while varying diversity pressure, holding diversity approximately fixed while varying population size, replacing DHD generation with matched-compute random valid search, and removing PDC metacontrol while retaining the same generators.

The two crucial experiments are:

$$N \text{ fixed}, \text{vary diversity pressure},$$

and

$$\text{diversity approximately fixed}, \text{vary } N.$$

If performance follows population size but not structural diversity, the central DHD claim is weakened.

14.5 Kill criteria

DHD should be reduced to an engineering subsystem rather than maintained as an independent scientific framework if any of the following persist across appropriately powered tests:

matched-compute random valid search matches DHD on hypothesis recall and validated discovery lead;

any apparent advantage vanishes when population size and model-family access are controlled;

benefits vanish under an appropriate adaptive multiple-testing procedure;

measured novelty is mostly syntactic rather than predictive, causal, temporal, or interventional;

archive metrics fail to predict useful later discovery;

one generator dominates universally and the remaining architecture contributes no measurable value;

PDC metacontrol adds no advantage over fixed or random operator allocation;

provenance cannot reproduce claimed early hypotheses;

causal/reflexive generators fail in synthetic environments where those structures are ground truth;

compute, memory, or latency overhead exceeds the value of any validated discovery advantage.

The Round-2 research packet contained one draft kill criterion based on “lineage-root multiplicity correction,” while the same theorem audit rejected lineage-root counting as a valid multiplicity procedure. This specification resolves

the inconsistency in favor of the theorem audit: the requirement is a statistically valid adaptive multiple-testing procedure, not root counting.

15. Formal status of major claims

The following table separates mathematics from design choices.

Statement	Status
typed coproduct H = coproduct_k H_k	definition / canonical candidate
multi-view comparison rather than universal scalar metric	design theorem-result of audit; canonical architecture
generator as typed partial stochastic map	definition
operator-path dependence	mathematical possibility; empirically testable relevance
epsilon-packing archive bound	proposition under total boundedness
greedy (1 - 1/e) for monotone submodular cardinality selection	imported theorem [10]
DPP determinant encourages feature-space diversity	established under PSD kernel [9]
DPP automatically compresses observational equivalence	false in general
global DHD ergodicity	open conjecture; not required
lineage roots equal independent statistical tests	rejected
controlled diversity improves discovery lead	empirical hypothesis
PDC metacontrol improves operator allocation	empirical hypothesis
DHD is empirically superior to novelty search/QD	unproven

16. Novelty and relation to prior art

DHD should not claim novelty for ensembles, novelty search, quality-diversity archives, DPPs, SMC populations, causal discovery, counterfactual reasoning, probabilistic programming, performative prediction, or bandit allocation. Those are established ideas.

The potentially distinctive contribution is the conjunction of five design commitments:

a typed heterogeneous archive whose members are retained without epistemic commitment;

external metacontrol that separates operator class, mutation magnitude, resource budget, and archive breadth;

a generator family explicitly designed to cover different divergence channels rather than only parameter-space neighborhoods;

persistent provenance integrated with model-space exploration and prospective registration;

an empirical test of controlled structural diversity under compute-matched and multiplicity-controlled conditions.

Whether that conjunction is sufficiently novel to constitute a distinct scientific framework is itself testable. If a simpler established novelty/QD system with the same model families and compute performs equivalently, DHD's contribution reduces to an engineering integration pattern.

17. Open problems

Eight problems remain intentionally open.

Heterogeneous PSD kernels. No universal positive-semidefinite kernel has been established across the full coproduct while preserving all causal, temporal, and predictive semantics.

Reachability. No concrete generator library has been proved complete even for a useful restricted model space.

Empirical advantage. Controlled structural diversity has not yet shown a compute-matched discovery advantage.

Adaptive multiplicity. The correct downstream inferential procedure for hypotheses generated adaptively from prior data and lineage remains unresolved.

Lineage dependence. Provenance may expose useful dependence structure, but no theorem converts genealogy into FDR/FWER control.

Resource-rational value. A principled value function over vector compute/memory/latency budgets remains to be developed.

Scalable contradiction management. The architecture is clear, but efficient contradiction indexing across very large heterogeneous archives is open.

Behavior descriptor design. Coverage is meaningful only relative to a declared descriptor space, yet a poor descriptor can hide valuable structural diversity or smuggle in epistemic quality.

18. Minimum viable DHD

The smallest implementation capable of testing Empirical Hypothesis 1 consists of:

a typed coproduct hypothesis space and executable interface contract;

three to five generator classes spanning local, regime/temporal, causal, model-class, and reflexive moves;

the hard validity/testability gate;

a thresholded novelty/coverage archive under explicit budget;

immutable provenance DAGs;

external control tuple u_t^D from PDC;

export to a separate adjudicator.

DPP selection, analogical generation, operator bandits, advanced causal kernels, and dormancy/resurrection are extensions rather than prerequisites. This minimum architecture should be tested first because a theory that only works after adding every optional mechanism is difficult to falsify.

19. Conclusion

Divergent Hypothesis Dynamics formalizes a narrow but important idea: an intelligent system should sometimes spend computation generating explanations that it does not currently believe. The framework is designed for situations in which model-family misspecification, regime change, causal ambiguity, temporal aliasing, or endogenous feedback makes local inference insufficient. Its core is not irrationality. It is **controlled exploration of explanatory structure**.

The mathematically defensible DHD architecture is a typed heterogeneous hypothesis space, a set of partial stochastic generators, a multi-view comparison layer, a bounded novelty/coverage archive, explicit contradiction and equivalence relations, immutable provenance, and an external metacontrol interface. Several attractive ideas from the first research round do not survive: the scalar exploration objective, a single divergence knob, an unconditional global distance, DPPs as the universal archive mechanism, automatic multiplicity correction from lineage, and unqualified ergodicity claims.

The architecture deliberately stops before belief. A later adjudicator must determine which hypotheses deserve epistemic weight, and a later policy layer must determine what to do. That separation is the scientific wager of DHD. If controlled structural divergence produces earlier validated discovery under matched resources and strict false-discovery control, DHD earns its place as an independent cognitive framework. If it does not, the kill criteria require reducing it to the simpler search mechanisms it resembles.

Appendix A. Principal notation

Symbol	Meaning
K	model-family index set
H_k	hypothesis family
H	typed coproduct hypothesis space
h	one executable hypothesis
M	executable generative model/program
Omega	predictive-consequence operator
Pi	provenance record
d_profile	multi-view distance/profile
G_g	typed generator
G	generator library
E_t	generation evidence/context
u_t^D	DHD control tuple from PDC
pi_t^G	generator-mixture policy
s_t	operator-specific mutation scale vector
B_t	vector resource budget

Symbol	Meaning
ell_t	archive novelty/admission threshold
C_t	candidate pool
A_t	active archive
D_t	dormant archive
Q_t	speculative quarantine
E_t^eq	equivalence/compression index (when ambiguity with evidence matters)
G_t^H	provenance DAG
beta_t	behavior descriptor map
nu(h|A)	behavioral novelty
Cov_epsilon(A)	declared behavioral coverage

Appendix B. Reference interface schema

A language-neutral hypothesis record should minimally expose:

```
Hypothesis                                                                {
                    id:          immutable          content/version          identifier
                                                 family:                    ModelFamily
                        executable_model_ref:            artifact            identifier
                                 variables:                typed               schema
                                       assumptions:                  list[Assumption]
                                    temporal_semantics:                 TemporalSpec
                                 predictive_interface:               PredictDistribution
                                     intervention_interface?:                Intervene
                                 counterfactual_interface?:               Counterfactual
          testability_class:     T1     |     T2     |     T3     |     T4     |     T5
                                  implications:                list[TestableImplication]
                               identification_status:               IdentificationStatus
                                     provenance_ref:                  ProvenanceNodeID
                                      generation_cost:                  ResourceVector
                               equivalence_tags:                list[EquivalenceClassRef]
                            internal_uncertainty?:               ModelInternalUncertainty
}
```

The record intentionally contains no belief_weight, confidence_from_DHD, TradeIntent, direction, size, order, or execution field.

Appendix C. Theorem and claim audit

Claim from research stage	Final disposition
scalar J_DHD = alpha N + beta D + gamma U + delta C - eta R	rejected: arbitrary commensuration and population inflation
scalar rho^D as universal divergence control	rejected: under-parameterized
weighted global d_H as canonical metric	rejected: multi-view profile retained
Gaussian kernel from arbitrary d is PSD	false in general; negative-type conditions required
DPP primary archive	optional candidate-subset mechanism only
DPP greedy selection “eliminates redundancy”	overstated; kernel-dependent and approximate
submodular greedy (1 - 1/e)	retained under theorem assumptions
archive size bounded by epsilon packing	retained under total-boundedness assumption
ordinary SMC degeneracy warning	retained with conditions; DHD is not a posterior SMC process
Markov equivalence means full observational identity	rejected without Markov/faithfulness/no-latent qualifications
lineage roots equal independent tests	rejected
global ergodicity of DHD	open; not required
structural diversity improves validated discovery	empirical hypothesis

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